Harvard University
Practice Questions for Physics 141 Final Exam

1. In this problem, you will explore Fourier transforms and study how biological systems apply filters to stimuli. To start, let's define some notation for the continuous case.

   - Let $x(t)$ and $h(t)$ denote two continuous time-series.
   - Let $X(\omega)$ and $H(\omega)$ denote their Fourier transforms. To be clear, we can write

   $$X(\omega) = \mathcal{F}(x)(\omega) = \int_{-\infty}^{\infty} dt\, x(t)e^{-i\omega t}$$

   - Define the convolution of two continuous time-series as

   $$(x * h)(t) = \int_{-\infty}^{\infty} d\tau\, x(\tau)h(t-\tau)$$

   (a) Prove the Fourier convolution theorem. That is, show that if $y = (x * h)$, then $Y(\omega) = X(\omega)H(\omega)$.

   Some receptor/signaling pathways that bacteria use to detect ligands act as low-pass filters. In an experiment, when you measure the response of bacteria to a ligand time course, your measurement will be a convolution of the true ligand time course with an impulse response. Model the impulse response $h(t)$ with an exponential function

   $$h(t) = \frac{1}{\tau}e^{-\frac{t}{\tau}}u(t)$$

   Where $\tau > 0$ and u(t) is the Heaviside step function ($u(t) = 1$ for $t \geq 0$ and 0 otherwise). Let us model the true ligand time course with a sinusoid

   $$x(t) = A\cos(\omega_0 t)$$

   Where $A > 0$. The bacteria sensing module is driven with a sinusoid of frequency ω_0. As we stated above, the output $y(t)$ (what we measure) is given by the convolution

   $$y(t) = (x * h)(t)$$

(b) Show that the frequency response $H(\omega)$ is given by

$$H(\omega) = \frac{1}{1 + i\omega\tau}$$

(c) Show that $x(t)$ has the Fourier transform

$$X(\omega) = A\pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))$$

(d) Using the convolution theorem that you proved in (a), find $Y(\omega)$.

$$y(t) = A\Re(H(\omega_0)e^{i\omega_0 t})$$

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Note that $H(-\omega_0) = (H(\omega_0))^*$ (* denotes complex conjugation). Note that $y(t)$ is a real number. A trick to solving this problem is to use the fact that for a complex number c,

$$c + c^* = 2\Re(c)$$

Where $\Re$ denotes taking the real part (i.e. if $c = a + bi$, $\Re(c) = a$).

(e) Define the gain (the amplitude response) as $G(\omega) = |H(\omega)| = \sqrt{H(\omega)(H(\omega))^*}$, where $(H(\omega))^*$ denotes the complex conjugate of $H(\omega)$. Find the gain G.

(f) Define the phase $\phi(\omega)$ as the argument of $H(\omega)$. Find the phase ϕ.

(g) Rewrite $y(t)$ in terms of the gain $G(\omega_0)$ and phase $\phi(\omega_0)$. You should be able to show that

$$y(t) = AG\cos(\omega_0 t + \phi)$$

(h) In the limits of $\omega_0 \ll \frac{1}{\tau}$ and $\omega_0 \gg \frac{1}{\tau}$, what is the gain and phase? Provide a biological interpretation of this result (how does your measurement/the receptor track fast/slow changes in the ligand concentration).

2. In the following problem, you will explore the statistics of the two-alternative forced choice (2AFC) task in Tinsley et al., 2016 (we studied this paper in class on Thursday). The paper reports $n = 2420$ single-photon events that passed post-selection, with an average proportion of correct response to be $\hat{p} = 0.516 \pm 0.010$. They also report $n = 242$ high-confidence responses, for which the proportion of correct responses was $\hat{p} = 0.60 \pm 0.03$. Note that in this notation $\hat{p}$ denotes the observed number of correct responses in a dataset and is an estimate of the probability of a correct response p. We are going to study, via a simple model, whether this data provides evidence that the true performance is above chance.

   If each trial is independent with a probability of p to be correct, then the number of correct responses X after n trials follows a binomial distribution. The sample proportion is defined as $\hat{p} = \frac{X}{n}$. For large n, $\hat{p}$ follows a normal distribution with mean p and variance $p(1-p)/n$.

(a) Explain why the baseline probability of a correct response for the 2AFC task in Tinsley et al., 2016 is $p_0 = 0.5$.

(b) Under a null hypothesis H_0, we assume that the probability of a correct response is $p_0 = 0.5$. For the $n = 2420$ trials and under this null hypothesis, what would you expect the mean μ_0 and standard deviation σ_0 of the data to be (in other words, what would the dataset look like if the baseline was true)?

(c) Compute the z-score for the $n = 2420$ case, which is given by

$$z = \frac{\hat{p} - p_0}{\sigma_0}$$

Explain what the z-score measures.

(d) Using a Z-test, compute the one-sided p-value by integrating directly. Explain what this p-value measures.

(e) Repeat the calculations in (b) to (d) for the $n = 242$ high-confidence responses.

3. Say we have a system with N particles and m different states, each with energies ϵ_i. Assume conservation of particle number and particle energy.

(a) Explain why

$$W = \frac{N!}{\prod_{s=1}^{m} n_s}$$

represents the number of ways each possible configuration (number of particles in each state) can be formed. What constraint do we have on the n_s?

(b) Explain why the real distribution of energies is the one with maximum W.

(c) Use Lagrange multipliers to maximize $\log W$ subject to the constraints of constant N and total energy E. Use Stirling's approximation

$$\ln n! \approx n \log n - n$$

Show that

$$n_s = Ae^{-B\varepsilon_s}$$

Do not solve for A and B.

(d) Recall that Shannon entropy for a discrete distribution is defined as

$$H = \sum p(x) \ln p(x)$$

Explain the utility of a maximum entropy distribution.

(e) Use Lagrange multipliers to rederive your answer from part (c) using maximum entropy.

4. (a) Write the definition of a Poisson distribution (1-2 sentences). Is it discrete or continuous?

(b) Recall that the Shannon entropy is given by

$$H = \sum p(x) \ln p(x)$$

We can add a factor $m(x)$:

$$H = \sum p(x) \ln \frac{p(x)}{m(x)}$$

if we have a nonuniform distribution of states. Let $p(k)$ define the probability of k events in an interval of length 1. How many ways can k events occur if we treat all order of events in the same way? Then, what is m?

(c) What are the two constraints we can put on $p(k)$?

(d) Recall our method of using Lagrangian multipliers to find the maximum entropy distribution. What would be the Lagrangian here?

(e) Use variational calculus to rederive the Poisson distribution.

5. An *E. coli* cell moves in 1D by alternating between straight runs at speed v (in the positive (+) or negative (-) direction) and instantaneous tumbles that can change its direction. Each run lasts for a random time with mean τ. After each tumble, the cell moves to the left or right with equal probability. Model tumbles as a Poisson process with rate $\lambda = \frac{1}{\tau}$. Let $x(t)$ denote the position of the bacteria and $u(t)$ its velocity.

   (a) Show that

   $$\langle x^2(t) \rangle = 2 \int_0^t ds\, (t-s) \langle u(s) u(0) \rangle$$

   (b) Show that $\langle u(t)u(0) \rangle = v^2 e^{-\frac{t}{\tau}}$. Hint: use the law of total expectation. Consider $\langle u(t)u(0)|A \rangle$ and $\langle u(t)u(0)|A^c \rangle$, where A is the event that there is no tumble between times 0 and t and A^c is the event that there is at least one tumble between times 0 and t.

   (c) Show that for $t \gg \tau$, $\langle x^2(t) \rangle \approx 2Dt$, where

   $$D = \frac{v^2 \tau}{2}$$